

A Unified Intelligence Layer for Software Supply Chain Governance

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Submission for double-blind review — SCORED '26

ABSTRACT

Artifacts such as Software Bills of Materials (SBOM), Vulnerability Exploitability eXchange (VEX), and Third-Party Notices (TPN) have advanced transparency but remain insufficient for consistent, scalable, and auditable release decision-making. These artifacts answer narrow, format-specific questions; none provides a decision-ready abstraction for determining release eligibility under combined security, legal, architectural, and AI-risk constraints. We present the Unified Intelligence Layer (UIL), a governance abstraction that transforms heterogeneous supply chain artifacts into structured, machine-interpretable assertions over which policy-as-code can be evaluated. We introduce the concept of *safety relevance* and position UIL as the layer that computes, attaches, and audits safety-relevance claims across five artifact classes. We systematize five complementary artifact classes (SBOM, VEX, TPN, Threat Model Bill of Materials (TMBOM), and AI Vulnerability Exchange (AIVEX)) into a single decision model with an explicit assertion schema, a normalization-and-correlation pipeline, and an auditable evaluation trail. We demonstrate feasibility through a proof-of-concept implementation on real CycloneDX 1.5 and OpenVEX 0.2.0 inputs; characterize scalability on synthetic inputs up to 50,000 components; and report a complementary experiment on seven real production SBOMs (3,099 components). UIL is presented as a governance abstraction rather than a standardized framework: it clarifies what a decision-ready supply chain layer must compute, regardless of which underlying tools produce the input artifacts.

CCS Concepts: Security and privacy -> Software security engineering; Software and application security. Software and its engineering -> Software verification and validation.

Keywords: software supply chain security, SBOM, VEX, TPN, AIVEX, TMBOM, policy-as-code, compliance automation, release governance, risk intelligence, decision governance

1 INTRODUCTION

A modern software release is governed by a half-dozen heterogeneous artifacts (SBOMs, VEX documents, Third-Party Notices, threat models, and AI risk descriptors), each produced by a different tool and owned by a different organizational function. No artifact, and no existing combination of them, yields a unified release decision: a human release manager must still reconcile them manually, introducing inconsistency, audit gaps, and delayed delivery cycles. UIL closes this loop by consuming all five artifact classes, correlating their signals across security, legal, architectural, and AI-risk dimensions, and emitting typed *safety-relevance assertions* (small, machine-evaluable claims like `NON_EXPLOITABLE` or `LICENSE_REVIEW_REQUIRED`) over which a policy engine (e.g., OPA/Rego) can produce a deterministic, auditable release decision. The contribution is the abstraction and the safety-relevance concept; we also include a working prototype, a scalability study to 50,000 components, and an experiment over 3,099 components drawn from seven real production SBOMs.

Software supply chain security has evolved rapidly under the combined pressure of pervasive open-source reuse, third-party dependency growth, and the recent emergence of AI-assisted software generation. Regulatory action, notably U.S. Executive Order 14028 [1] and the European Union Cyber Resilience Act [2], has elevated software transparency from a best-practice concern to a procurement and market-access requirement. Organizations now produce a growing and heterogeneous set of compliance and security artifacts: SBOMs [3], VEX documents [4], Third-Party Notices [5], threat-model documentation, and emerging AI/ML risk descriptors.

Despite this artifact proliferation, software release governance remains largely manual, fragmented, and inconsistent. Common consequences observed in release practice include:

- inconsistent release decisions across products and business units;
- duplicated analysis effort across security, legal, and engineering reviews;
- weak audit traceability between source evidence and final release decisions;

- delayed delivery cycles caused by serial, human-mediated reviews.

This paper makes four contributions. First, we systematize the current software supply chain artifact landscape into five complementary classes and identify five structural gaps that prevent these artifacts from supporting automated release governance (§2–§3). Second, we introduce the concept of UIL, a governance abstraction that computes, attaches, and audits safety-relevance claims through structured assertions suitable for policy-as-code evaluation (§4–§6). Third, we extend the abstraction to AI systems through TMBOM and AIVEX, and compare UIL against adjacent frameworks including SLSA [6], in-toto [7], GUAC [8], and S2C2F [9] (§7–§9). Fourth, we demonstrate feasibility through a proof-of-concept implementation (§8.1–§8.4). We close with limitations, threats to validity, and open research problems (§10–§11).

Scope. UIL is positioned as a governance abstraction, not a new artifact format, a new SBOM dialect, or a competing build-integrity framework. It assumes the existence of SBOM, VEX, and similar artifacts and concerns itself with how to convert their combined signal into a release decision that is consistent, explainable, and auditable. UIL is complementary to provenance-and-integrity frameworks such as SLSA and in-toto: those frameworks attest *how* an artifact was built; UIL evaluates *whether* the resulting artifact may be released.

2 PROBLEM STATEMENT

We frame the governance problem as a gap between artifact visibility and decision intelligence. Current practice exhibits three structural deficiencies and two operational ones.

2.1 Structural Deficiencies

- **Artifact Fragmentation.** SBOMs, VEX, and TPNs are owned by different organizational functions and stored in separate systems; reconciliation is largely manual.
- **Absent Semantic Correlation.** Existing schemas focus on specific domains and provide no widely adopted mechanism for policy-driven cross-domain correlation; today this is done in spreadsheets or non-interoperable knowledge graphs.
- **No Decision-Ready Abstraction.** Release eligibility is not inherent in any single artifact; current systems cannot evaluate release policies against raw artifacts without first deriving contextual claims.

2.2 Operational Consequences

- **Inconsistent Release Decisions.** Without explicit policy models, release decisions vary across teams for reasons unrelated to genuine risk differences.
- **Brittle Audit Trails.** Organizations struggle to maintain a reproducible chain of evidence linking artifacts to release decisions.

These five gaps share a root cause: existing artifacts provide *visibility* but not *decision intelligence*. A governance layer operating above them, one that normalizes, correlates, and reasons over their combined signal, is the missing structural component. §3 surveys the five artifact classes; §4 introduces UIL as that layer.

3 SOFTWARE SUPPLY CHAIN ARTIFACT LANDSCAPE

Modern supply chains produce specialized artifacts, each answering a distinct question. We organize them into five classes; Table 1 summarizes their scope and limitations.

Class	Answers	Primary Limitation
SBOM	What components are present?	No risk semantics
VEX	Which vulns are exploitable?	Security-only scope
TPN	What license obligations apply?	Weakly standardized
TMBOM	Where are trust boundaries?	Not widely standardized
AIVEX	What AI/ML risks apply?	Schema actively standardizing in CycloneDX 2.0 [19, 20]

Table 1. The five artifact classes covered by UIL.

3.1 SBOM — Software Bill of Materials

An SBOM is a nested inventory of the components and dependencies that make up a software artifact, typically expressed in SPDX [10] or CycloneDX [11]. SBOMs answer: what components are present? They are the foundation of modern

supply-chain transparency, but they do not indicate whether any of those components introduce unacceptable risk under the consuming organization's policy.

3.2 VEX — Vulnerability Exploitability eXchange

VEX documents [4], expressible in CSAF, CycloneDX-VEX, or OpenVEX [12], assert the exploitability status of a known vulnerability in a specific product context. VEX makes vulnerability response tractable by suppressing irrelevant findings, but it does not encode legal, architectural, or AI-specific risk.

3.3 TPN — Third-Party Notices

Third-Party Notices capture component-level license terms, attribution requirements, and copyleft obligations. TPN remains weakly standardized and is rarely correlated with security signals at decision time [13, 14, 15].

3.4 TMBOM — Threat Model Bill of Materials

TMBOM (Threat Model Bill of Materials) captures system design assumptions, trust boundaries, data flows, and applied mitigations. It converts a finding from 'CVE exists in X' into 'CVE is unreachable because X sits behind trust boundary Y.'

3.5 AIVEX — AI Vulnerability Exchange

AIVEX extends the VEX pattern to AI/ML components, covering model behavioral risks, inference-time attack surfaces (prompt injection per CWE-1427, jailbreaks), and adversarial robustness. The CycloneDX specification community is standardizing AIVEX in CycloneDX 2.0 [19, 20]; UIL treats AIVEX as a logical role above this emerging standard.

4 THE UNIFIED INTELLIGENCE LAYER

4.1 The Governance Problem UIL Solves

UIL is designed to compute a single property across five artifact classes: whether a given component, vulnerability, license obligation, architectural exposure, or AI behavior is relevant to the safe and policy-compliant release of a software product into a specific deployment context. No single artifact carries this property; it emerges from the correlation of all five in deployment context. UIL is designed to compute, attach, and audit that correlated signal as structured, machine-evaluable assertions. Formally, for a component c in deployment context C , UIL computes:

$$SR(c, C) = \text{Phi_P} (\text{sig_S}(c,C), \text{sig_L}(c,C), \text{sig_A}(c,C), \text{sig_AI}(c,C))$$

where sig_S , sig_L , sig_A , sig_AI are signal-extraction functions over security (SBOM+VEX), licensing (TPN), architectural exposure (TMBOM), and AI behavior (AIVEX) artifacts respectively, each emitting typed claims; and Phi_P is the aggregation function induced by the organization's release policy. The SR function carries no hidden state: given the same artifacts and policy version, it yields the same claim, evidence chain, and confidence. Adding a documented mitigation to TMBOM or AIVEX may only reduce or preserve claim severity, never increase it. Two organizations with different release policies see the same sig sub-claims and diverge only at Phi_P , which is the property that makes UIL portable across governance regimes.

Safety relevance in UIL is explicitly policy-dependent: SR is parameterized by Phi_P , and two organizations applying different policies to the same artifacts will produce different claim assignments. This is a design choice — safety relevance in practice is always contextual and organizational, and UIL makes that context-dependence auditable rather than hiding it in informal judgment. Two policies can produce contradictory relevance conclusions for the same component if their TMBOM trust boundaries differ; UIL's policy-aggregation separation ensures these divergences are traceable to the specific Phi_P version. Safety relevance in UIL is ordinal rather than binary: the claim vocabulary of §5.2 defines a partial order from NON_EXPLOITABLE through REMEDIATION_REQUIRED, with the confidence model adding a continuous dimension orthogonal to the claim tier.

4.2 The Layer

We define UIL as a governance abstraction that operates above the artifact layer and performs four functions:

- **Normalization.** Heterogeneous artifact formats (SPDX vs. CycloneDX, CSAF vs. OpenVEX, prose TPN vs. structured legal data) are mapped into a common internal representation.
- **Identity resolution.** Components are resolved to canonical identities across artifacts, addressing the well-known PURL/CPE/SWID-tag mismatch and version-range ambiguity that breaks naïve correlation.

- **Correlation.** Security, legal, architectural, and AI-risk signals about the same component are joined, weighted, and combined into safety-relevance context.
- **Assertion.** The correlated state is emitted as structured, machine-evaluable safety-relevance claims over which policy-as-code can be evaluated.

4.3 Pipeline Architecture

Heterogeneous artifacts enter and pass through five stages: (1) Normalization maps SPDX, CycloneDX, CSAF, OpenVEX, and prose TPN inputs into a common internal representation; (2) Identity Resolution unifies PURL, CPE, SWID, and content-digest identifiers into canonical component identities; (3) the Correlation Engine extracts typed sub-claims via the signal functions sig_S, sig_L, sig_A, and sig_AI; (4) the Assertion Store records the resulting safety-relevance tuples with evidence pointers and confidence; and (5) the Policy Engine evaluates Phi_P over those assertions to produce release decisions. The Assertion Store is independently queryable, which gives the audit trail its reproducibility property.

Algorithm 1 specifies the hierarchical identity resolution procedure used in the §8.1 prototype:

```

Input: reference r (PURL, CPE, SWID, or digest)
       normalized component set C, optional digest d
Output: (component, strategy, q_identity)

1: if d provided: for each c in C with c.digest defined:
2:   if c.digest == d -> return (c, DIGEST, 1.00)
3: for each c in C:
4:   if c.canonical_id == r -> return (c, EXACT_PURL, 0.90)
5:   coords_r := normalize(r) // canonical (type, name, resolved version)
6:   if coords_r != |_|_:
7:     if normalize(c.canonical_id) == coords_r
8:       -> return (c, NORMALIZED_COORDS, 0.75)
9: for each c in C:
10:  if c.type == coords_r.type and c.name == coords_r.name
11:    -> return (c, FUZZY_VERSION, 0.50)
12: return (null, NONE, 0.00)

```

The strategy returned by Algorithm 1 directly determines Q_identity for downstream assertions. The values are anchored to SLSA provenance levels [6]: DIGEST=L3, EXACT_PURL=L2, NORMALIZED_COORDS=L1, FUZZY_VERSION=unverified. The geometric spacing preserves monotonicity and ensures each tier produces meaningfully different policy outcomes at the 0.40 threshold. Empirical calibration against labeled resolution outcomes is identified as future work. A key edge case is cross-namespace aliasing (e.g., GitHub URL in SBOM vs. legacy CPE in VEX). UIL handles this in Stage 1 via a synonym map (CPE-to-PURL from NVD/OSV) before Algorithm 1 executes. One-to-many mappings emit one assertion per candidate with the lowest Q_identity propagated; unresolvable mismatches return NONE (Q_identity=0.00), suppressing the assertion and flagging it for out-of-band resolution. Cross-ecosystem bridging remains an open problem (§11).

Strategy	Q_identity	When used
DIGEST	1.00	Content-addressed digest match
EXACT_PURL	0.90	Canonical PURL exact match
NORMALIZED_COORDS	0.75	Type + name + resolved version match
FUZZY_VERSION	0.50	Type + name match, version range overlap
NONE	0.00	No match; assertion suppressed

Table 2. Identity confidence scores by resolution strategy.

5 ASSERTION MODEL

5.1 Definition

An assertion is a tuple: (*component*, *context*, *claim*, *evidence*, *confidence*, *timestamp*)

- **component:** Canonical identity (e.g., PURL or content-addressed digest).

- **context:** Deployment or release scope under which the claim is evaluated.
- **claim:** Typed claim from the closed vocabulary of §5.2 (NON_EXPLOITABLE, LICENSE_COMPLIANT, ELEVATED_THREAT, AI_BEHAVIOR_RISK, etc.).
- **evidence:** Referential pointers to source artifacts justifying the claim (VEX statement, TPN clause, TMBOM node, AIVEX entry).
- **confidence:** $\text{Conf}(A) = Q_{\text{artifact}} \times Q_{\text{identity}} \times Q_{\text{correlation}}$. Q_{artifact} is drawn from a provenance lookup (signed CycloneDX 1.5 = 0.95, unsigned = 0.85, CycloneDX 1.2 = 0.70; SLSA levels [6] provide a principled basis). $Q_{\text{correlation}}$ reflects inferential strength: direct VEX = 0.95, inferred reachability = 0.55. Multiplicative composition (e.g., $0.95 \times 0.90 \times 0.95 = 0.81$) correctly reflects compounded uncertainty. A single degraded input can drop confidence substantially: $Q_{\text{artifact}}=0.70$ (CycloneDX 1.2 SBOM) yields $0.70 \times 0.90 \times 0.95 = 0.60$, potentially triggering unnecessary manual review. Organizations may substitute a weighted average for less dampening, or a weakest-link minimum for stricter governance; per-claim thresholds tune the false-block/missed-risk trade-off. Low-confidence claims route to human review.
- **timestamp:** Evaluation time; relevant because exploitability and AI risks evolve.

5.2 Claim Vocabulary

Table 3 lists the core claim types that UIL emits across the five artifact classes.

Claim	Dimension	Meaning
NON_EXPLOITABLE	Security (sig_S)	Vulnerability present but suppressed by VEX context or TMBOM isolation
REMEDATION_REQUIRED	Security (sig_S)	Exploitable finding requiring patch or mitigation before release
LICENSE_COMPLIANT	Licensing (sig_L)	Component meets license obligations under the given distribution context
LICENSE_REVIEW_REQUIRED	Licensing (sig_L)	Copyright or attribution obligation requires legal review or remediation
ELEVATED_THREAT	Architecture (sig_A)	Component sits on a trust boundary adjacent to a sensitive data path in the deployment context, as documented in TMBOM
AI_BEHAVIOR_RISK	AI (sig_AI)	Model exhibits adversarial-input or behavioral risk relevant to deployment use
PROMPT_INJECTION_RISK_BOUNDED	AI (sig_AI)	Prompt-injection susceptibility (CWE-1427) mitigated by TMBOM-defined input controls

Table 3. Core safety-relevance claim vocabulary emitted by UIL.

The PROMPT_INJECTION_RISK_BOUNDED claim carries an inherently higher verification burden than security or licensing claims: input-control adequacy is an organizational assertion. UIL treats this claim as requiring *double evidence pointers* (both an AIVEX entry documenting the susceptibility and a TMBOM mitigation node documenting the control) and the prototype enforces this in the correlation engine by requiring both sig_AI and sig_A evidence before emitting the claim with confidence above 0.5.

5.3 Example Assertions

- *openssl@3.0.7* in context "enclave-isolated TLS terminator" has claim NON_EXPLOITABLE for CVE-2023-XXXX, evidenced by VEX statement V-1023 and TMBOM trust-boundary B-04, with high confidence.
- *lib-foo@2.1* in context "binary distribution to external customers" has claim LICENSE_REVIEW_REQUIRED under copyleft obligations, evidenced by TPN clause T-77, with high confidence.
- *vendor-sdk@4.2* in context "network-facing service" has claim ELEVATED_THREAT under current deployment topology, evidenced by TMBOM exposure node E-12, with medium confidence.
- *llm-component-X* in context "customer-facing chat" has claim PROMPT_INJECTION_RISK_BOUNDED (CWE-1427) under defined input class, evidenced by AIVEX entry A-09 and TMBOM mitigation M-03, with medium confidence.

6 POLICY-DRIVEN RELEASE GOVERNANCE

Once assertions exist, release governance reduces to evaluating a set of policy rules over those assertions. We adopt the policy-as-code model familiar from OPA/Rego [17]. A policy decision over a release R is the result of evaluating every applicable rule over the assertions associated with R. Rules can:

- approve a release automatically when all applicable assertions clear policy thresholds;
- block a release when one or more assertions cross block thresholds;
- require remediation, holding release pending an updated assertion;
- require formal risk acceptance, routing to a designated approver.

Listing 1 shows an excerpt from the strict-veto Rego policy used by the §8.1 prototype:

```
package uil.release
min_confidence := 0.4 # assertions below threshold routed to human review
block_reasons[reason] {
  some a in input.assertions
  a.claim == "REMEDIATION_REQUIRED"
  reason := sprintf("block: %s", [a.component])
}
risk_acceptance_reasons[r] {
  some a in input.assertions
  a.claim == "ELEVATED_THREAT"
  r := a.component
}
decision := "BLOCK"          if { count(block_reasons) > 0 }
decision := "RISK_ACCEPT_REQ" if { count(block_reasons) == 0
                                count(risk_acceptance_reasons) > 0 }
decision := "APPROVE"        if { count(block_reasons) == 0
                                count(risk_acceptance_reasons) == 0 }
```

Listing 1. Excerpt from policy/release.rego (strict-veto Phi_P).

Three properties of this model are worth emphasizing. First, the policy operates over assertions, not over raw artifacts. Second, every decision is reproducible: given the same assertion set and the same policy version, the same decision is reached. Third, policy evaluation runs locally against the Assertion Store at each consumer's site; UIL specifies what each consumer must compute, not a central authority that computes it for them.

7 EXTENDING GOVERNANCE TO AI SYSTEMS

Traditional supply chain governance is insufficient for AI-driven systems. A model artifact behaves quite differently from a library: its risk profile depends on training-data provenance, fine-tuning history, and runtime input distributions in ways that no SBOM or VEX can capture. We extend the artifact landscape through two emerging constructs.

7.1 TMBOM

TMBOM captures trust boundaries, architectural mitigations, and threat exposure context. A CVE in an isolated component is a different release decision from the same CVE in an internet-facing component; TMBOM is the artifact that distinguishes them.

7.2 AIVEX

AIVEX extends the VEX pattern to AI/ML components. It captures model behavioral risks, training-data provenance concerns, inference-time attack surfaces (prompt injection as vehicle per CWE-1427, indirect injection, jailbreaks), and adversarial robustness measurements. Combined with TMBOM, AIVEX allows UIL to issue assertions over AI behavior in deployment context, for example asserting that a model is acceptable for an internal data-classification task but not for a customer-facing summarization task, on the basis of the same underlying model card but different TMBOM exposure.

The CycloneDX specification community is actively developing the schema substrate that AIVEX requires. CycloneDX issue #956 [19] establishes that AI behavioral risk assertions (prompt-injection exploitability, agent abuse, model behavior drift) map onto the CycloneDX 2.0 vulnerability and risk objects using CWE-1427 as the weakness anchor and MITRE ATLAS alongside ATT&CK; as the threat classification taxonomy. CycloneDX issue #333 [20] is developing a complementary

AI-specific evidence taxonomy for prompt transcripts, behavioral tests, and model-artifact integrity. The CycloneDX 2.0 RFC is scheduled for August-September 2026.

UIL's `sigma_AI` extractor is decoupled from CycloneDX serialization format. It operates over UIL's normalized AIVEX representation (component identity, risk type, CWE anchor, MITRE ATLAS reference, mitigation pointer) rather than raw JSON. If the CycloneDX 2.0 schema changes, only Stage 1's normalization adapter requires updating; the correlation logic, assertion vocabulary, and policy engine are unaffected — the same format-independence that allows UIL to consume SPDX and CycloneDX SBOMs interchangeably. Backward compatibility with CycloneDX 1.5/1.6 is maintained: the normalizer extracts AI risk signals from `modelCard.considerations` and `modelCard.properties`, mapping them to the same normalized AIVEX representation.

Together, TMBOM and AIVEX extend UIL beyond software composition into system behavior and operational risk intelligence, a scope traditional supply chain frameworks were never designed to cover.

8 WORKED EXAMPLE AND EVALUATION

8.1 Prototype

The prototype source code is publicly available at <https://anonymous.4open.science/r/uil-prototype-231C> under an Apache 2.0 license. An interactive browser-based demonstration is available at <https://anonymous.4open.science/r/uil-prototype-231C/docs/index.html> — it runs the full five-stage pipeline in the browser, allowing reviewers to inspect assertion generation, confidence computation, and policy evaluation interactively without installing any software.

A proof-of-concept implementation of UIL has been developed in Python and Rego demonstrating that the abstraction is executable end-to-end over real CycloneDX 1.5 SBOMs, OpenVEX 0.2.0 statements, and structured TPN, TMBOM, and AIVEX inputs. Stage 5 ships in two parallel forms: a Rego policy file and an equivalent Python evaluator. Running the prototype on a synthetic but format-conformant input set reproduces the worked example's outcome. Pipeline stages emit seven assertions: three security sub-claims (one `NON_EXPLOITABLE` for openssl cross-referencing VEX with TMBOM boundary B-04; two `REMEDIATION_REQUIRED` for CVE-affected components), two licensing sub-claims, one `ELEVATED_THREAT` from TMBOM, and one `PROMPT_INJECTION_RISK_BOUNDED` (CWE-1427) from AIVEX. The strict-veto `Phi_P` policy aggregates these into a single decision (`BLOCK`) with full evidence pointers. The decision is deterministic: rerunning against the same inputs and policy version produces the same output.

Listing 2 reproduces one assertion verbatim from the prototype's Assertion Store:

```
{
  "component": "pkg:generic/openssl@3.0.7",
  "context": "enclave-isolated TLS terminator + gateway",
  "claim": "NON_EXPLOITABLE",
  "dimension": "security",
  "evidence": ["VEX:.../release-r1#CVE-2023-XXXX", "TMBOM:B-04"],
  "confidence": { "q_artifact": 0.85, "q_identity": 0.90,
                  "q_correlation": 0.95, "overall": 0.727, "label": "high" },
  "timestamp": "2026-05-27T01:58:31Z",
  "detail": "CVE-2023-XXXX; TLS handshake path not invoked; boundary B-04."
}
```

Listing 2. One assertion from the prototype Assertion Store, verbatim.

The `sig` function implementations underlying UIL have received independent commercial and community validation: a production vendor implementation of the security and architectural signal extractors has shipped, and the maintainers of CycloneDX, SPDX, OpenSSF, and OpenChain have engaged substantively with the assertion vocabulary. UIL treats those validations as confirmation that the governance pipeline is implementable without changes to any core SBOM specification.

The UIL governance abstraction has received independent practitioner validation through two external venues: (1) a presentation at an ISACA chapter webinar [withheld for review] where security governance practitioners identified the visibility-to-decision-intelligence gap as a recognized operational pain point, confirming the governance problem of Section 2; and (2) acceptance of a Supply Chain Security track session at swampUP 2026 (JFrog annual software supply chain security conference, September 2026, New York) [withheld for review], where an independent program committee selected the

framework for a practitioner audience focused on enterprise SBOM governance workflows. These validations confirm practitioner resonance with the governance problem and abstraction without measuring assertion correctness.

8.2 Scalability

To characterize feasibility and scalability bounds independently of artifact-specific variation, Table 4 reports end-to-end pipeline performance on format-conformant synthetic inputs from 10 to 50,000 components, run twice at each size to verify rerun determinism. Accuracy metrics on real corpora are deferred to future work as identified in §11.

Components	Assertions	SBOM size	Runtime (s)	Policy (s)	RSS (MB)
10	6	2.9 KB	0.002	<0.001	19.2
100	34	26.4 KB	0.004	<0.001	19.6
1,000	337	265 KB	0.034	<0.001	23.3
10,000	3,328	2.6 MB	0.661	0.001	58.6
50,000	17,007	13.2 MB	11.110	0.002	227.4

Table 4. Prototype scalability on synthetic inputs. All sizes produce deterministic rerun output.

The pipeline scales to 50,000 components in approximately 11 seconds of single-threaded Python on commodity hardware. Peak resident memory remains under 230 MB. Policy evaluation is essentially free relative to the pipeline (2 ms over 17,007 assertions at N=50,000), confirming that cost concentrates in correlation rather than aggregation, which is the property that makes the policy-aggregation separation of §4.1 architecturally sound.

8.3 Real-SBOM Experiment

Seven publicly available CycloneDX SBOMs were retrieved from the official CycloneDX bom-examples corpus at commit a3f1c9e (accessed May 2026): CERN LHC resource-tooling SBOM, Dropwizard 1.3.15, OWASP Juice Shop v11.1.2, Keycloak 10.0.2, Laravel 7.12.0, Proton Bridge v1.6.3, and ProtonMail Webclient v4. Together these span 3,099 real components across Maven, npm, PyPI, Composer, and Go ecosystems. We overlaid a 4% synthetic-VEX coverage and a minimal TMBOM on each SBOM. Table 5 reports per-project results.

Project	Components	Lic. cov.	sig_S	sig_A	Runtime
CERN LHC VDM Editor	43	100%	1	1	<0.001s
Dropwizard 1.3.15	167	79.0%	7	1	<0.001s
OWASP Juice Shop v11.1.2	840	95.6%	37	1	0.001s
Keycloak 10.0.2	903	96.7%	43	1	0.001s
Laravel 7.12.0	62	100%	3	1	<0.001s
Proton Bridge v1.6.3	201	96.0%	4	1	<0.001s
ProtonMail Webclient v4	883	98.4%	37	1	0.001s
Aggregate (7 projects)	3,099	95.1%	132	7	—

Table 5. Real-SBOM experiment over seven production CycloneDX SBOMs.

The pipeline processed all 7 projects without modification. Identity resolution: 59% digest (Q_identity=1.00), 41% exact PURL (Q_identity=0.90) — all 3,099 components resolved to the two highest-confidence strategies, with zero FUZZY_VERSION or NONE outcomes, providing a partial accuracy signal. License coverage ranged 79.0%-100%, confirming artifact-quality heterogeneity (§11). Full precision/recall requires adjudicated ground-truth labels; identified as future work.

8.4 What Artifact-Only Workflows Cannot Produce

Under artifact-only review of the §8 scenario, four functions inspect four artifact streams in parallel. Each produces a correct artifact for its own dimension; none produces a release decision. What UIL produces that no artifact-only tool can emit, because it depends on signal from two artifacts owned by two functions, is the cross-dimensional assertion set: a NON_EXPLOITABLE assertion citing VEX V-1023 and TMBOM B-04 is not a fact any single artifact tool can emit.

Likewise for PROMPT_INJECTION_RISK_BOUNDED (CWE-1427) citing AIVEX A-09 and TMBOM M-03. These cross-dimensional assertions, with audit trails linking each to source evidence, are the new artifact UIL produces, and the artifact release governance has been doing without.

9 COMPARISON WITH RELATED FRAMEWORKS

SLSA [6] specifies build-integrity levels and the provenance attestations needed to achieve them. SLSA asks how the artifact was built; UIL asks whether it should be released. SLSA provenance is a valuable input to UIL assertions but is not itself a governance layer.

in-toto [7] provides a framework for cryptographically attesting steps in a software supply chain. UIL does not replicate in-toto attestations; instead, in-toto attestations can serve as evidence elements within UIL assertions.

GUAC [8] aggregates software supply chain metadata into a queryable knowledge graph. A GUAC knowledge graph is a natural Stage 1 substrate for UIL. The key distinction: GUAC answers metadata queries within a single artifact namespace; it cannot produce typed cross-artifact assertions. A NON_EXPLOITABLE claim citing both VEX and TMBOM depends on sig_S and sig_A simultaneously — no single-namespace graph query produces this output. UIL formalizes the cross-artifact inference that GUAC's query model does not express.

S2C2F [9] prescribes practices and maturity levels for consuming open-source dependencies. S2C2F is a practice framework; UIL is a decision abstraction. They operate at different levels and are compatible.

Sigstore [18] provides signing and transparency infrastructure for software artifacts. Sigstore signatures are evidence inputs to UIL, not competing layers.

Commercial SBOM/SCA platforms have begun incorporating VEX-aware risk prioritization and context-dependent exploitability reasoning. These implementations remain single-dimensional (security-only, via VEX-over-SBOM); to the best of our knowledge, no publicly documented implementation has yet exposed a unified cross-domain assertion model spanning security, license compliance, architecture (TMBOM), and AI behavior (AIVEX), evaluable against executable release policies. UIL is positioned at exactly that gap. Adjacent open-source tools such as Anchore Enterprise, Dependency-Track, and OpenSSF Scorecard occupy related niches in artifact aggregation and trust scoring and are likewise complementary rather than competing.

Structural gap (§2)	SLSA	in-toto	GUAC	Comm. SCA	UIL
Artifact fragmentation	No	No	Part	Part	Yes
Semantic correlation	No	No	Part	Part	Yes
Decision-ready abstraction	No	No	No	No	Yes
Policy-driven release	No	No	Part	Part	Yes
Reproducible audit trail	Part	Part	Part	No	Yes
AI/ML risk integration	No	No	No	Part	Yes

Table 6. Capability comparison against structural gaps (§2). Yes = addressed; Part = partial; No = not addressed.

What adjacent frameworks cannot express. UIL's assertion layer produces cross-dimensional claims that depend simultaneously on artifacts owned by different organizational functions. A NON_EXPLOITABLE assertion citing both VEX and TMBOM cannot be emitted by any single artifact tool; neither can PROMPT_INJECTION_RISK_BOUNDED, which requires both AIVEX and TMBOM evidence. SLSA, in-toto, GUAC, and SCA platforms all operate within a single artifact namespace. UIL's assertion vocabulary formalizes the cross-artifact inferences release managers currently perform in their heads, inconsistently, without an audit trail — the structural gap Table 6 encodes under decision-ready abstraction.

10 AUDITABILITY AND COMPLIANCE

Auditability is the design goal. The full governance pipeline — Artifacts > Assertions > Policy Evaluation > Decision — is fully recorded. An auditor can trace any release decision back to its policy version, assertion set, and source artifacts. Assertion Store outputs can be signed as in-toto [7] attestations binding the assertion set, policy, and decision into a tamper-evident, independently verifiable package, supporting EO 14028, CRA, HIPAA, and PCI compliance reporting.

11 LIMITATIONS AND OPEN PROBLEMS

- **Artifact-Quality Heterogeneity.** UIL's assertions depend entirely on input quality. Real-world data frequently suffers from incomplete SBOMs, prose-based TPNs, and partial VEX coverage, which degrades decision confidence.
- **Identity Resolution.** Cross-artifact component identification remains unsolved. Overlapping and non-coextensive namespaces (PURL, CPE, SWID, content digests) cause naïve matching to yield false positives and false negatives.
- **Inference Errors in Correlation.** Aggregating signals across security, legal, and AI dimensions introduces algorithmic logic that may itself be flawed, requiring explicit confidence models and manual overrides.
- **Unstandardized AI-Risk Representation.** The CycloneDX 2.0 AIVEX schema [19, 20] is under active development. No public corpus of structured TMBOM or AIVEX artifacts currently exists; UIL defines these as logical roles so the normalization adapter in Stage 1 can consume any structured representation as schemas stabilize.
- **Validation Scope.** The evaluation demonstrates pipeline feasibility and scalability, not assertion correctness. Validating correctness requires: (a) a labeled corpus with ground-truth exploitability and safety-relevance annotations adjudicated by domain experts; (b) human-expert policy-decision agreement studies; and (c) longitudinal deployment data linking assertions to realized outcomes. None of these corpora or studies currently exist in the public domain. Constructing them is the primary empirical research agenda for future work. The contribution of this paper is the governance abstraction and feasibility demonstration.
- **Sociotechnical Complexity.** Aligning traditionally isolated functions (security, legal, engineering, compliance) around a shared policy engine is a major operational hurdle.
- **Policy Authoring Gap.** Translating informal compliance policies into executable policy-as-code is a skill most organizations have yet to acquire. A practical transition path is a two-tier model: a central compliance team maintains the baseline `release.rego` encoding organizational-wide hard blocks (REMEDIATION_REQUIRED, LICENSE_REVIEW_REQUIRED), while individual product owners extend it with context-specific risk-acceptance rules and confidence thresholds for their deployment environments. This separates the policy authoring concern from the policy evaluation concern and reduces the Rego expertise required at the product level.

11.1 Threats to Validity

- **Construct validity.** TMBOM and AIVEX are emerging rather than standardized; the schemas exercised by the prototype are experimental. The SR aggregation function is defined operationally; alternative formalizations would yield different outputs.
- **External validity.** The scalability experiment uses synthetic inputs; the real-SBOM experiment covers seven open-source projects. New artifact classes (hardware-attestation, AI training-data provenance) may not fit cleanly into the sig structure.
- **Internal validity.** Algorithm 1's heuristics, the per-strategy $Q_{identity}$ values, and the multiplicative confidence composition are design choices supported by the prototype but not empirically calibrated.

11.2 Threat Model and Trust Assumptions

UIL operates above an input layer of supplier-authored artifacts, so input integrity is a first-order concern. Adversary capabilities include: a malicious supplier issuing self-serving VEX; an intermediary tampering with artifacts in transport; a typosquatted package whose PURL fuzzy-matches a legitimate identity; or a compromised internal tool generating malformed TMBOM or AIVEX. The policy engine and Assertion Store are assumed trusted; crypto primitives are unbroken.

Conflicting assertions and assessor authority. When a supplier VEX asserts `NON_EXPLOITABLE` but internal TMBOM indicates the component is internet-facing and exposed, UIL emits both assertions. Under strict-veto Φ_P , the `ELEVATED_THREAT` from TMBOM routes to risk acceptance regardless of the supplier VEX claim, because the TMBOM carries higher $Q_{artifact}$ (internal analysis > supplier self-attestation). A natural extension is $Q_{artifact} = Q_{source} \times Q_{provenance}$, where Q_{source} reflects the assessor authority tier and $Q_{provenance}$ reflects cryptographic attestation strength via SLSA [6] or Sigstore [18]; this is identified as priority future work.

UIL composes with Sigstore [18] (authorship integrity), in-toto [7] (build-step attestation), and SLSA [6] ($Q_{artifact}$ bounds). Content-trust attestations at Algorithm 1's digest tier eliminate fuzzy-fallback risk; issuer-reputation models remain open.

12 CONCLUSION

Existing supply chain artifacts provide visibility but not decision intelligence. UIL closes this gap by transforming fragmented artifacts into correlated, machine-interpretable assertions over which policy-as-code can be evaluated. By extending the same abstraction across TMBOM and AIVEX — and designing sig_AI to sit above the CycloneDX 2.0 AI behavioral risk schema currently under standardization [19, 20] — UIL covers traditional software and AI-driven systems within one model.

Whether the community standardizes on a single UIL-shaped layer or implements several complementary ones is a secondary question. The primary contribution of this paper is to make explicit what any such layer must compute: a normalized, correlated, policy-evaluable assertion over the combined safety-relevance signal of the five artifact classes. The prototype demonstrates that this computation is feasible, deterministic, and space-efficient at enterprise scale. The open problems identified in §11 (identity resolution accuracy, AI-risk schema stabilization, and real-world artifact corpora evaluation) define the research agenda for turning this governance abstraction into a production-grade standard.

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